Learning Objectives

At the end of this lesson you shall be able to

- Understand the purpose of sanitation.
- Explain the quatity and quality of sewage.
- Know about Construction and testing of sewers.
- Describe the various processes involved in Sewage treatment Plant.
- Know the principles and working of septic tank.
- Get an idea about solid waste management.
- Explain the effects and prevention of various types of pollutions.

5.1 Introduction

Sanitary Engineering is the application of engineering method to improve sanitation of human communities. It is achieved by the removal and disposal of human waste to promote safe and healthy environment.

5.1.1 Purpose of Sanitation

1. To maintain safe environment.
2. Proper disposal of human excreta and waste water.
3. To avoid pollution of soil and water.
4. Reuse and recycling of waste and waste water.

5.1.2 Terms Used in Sanitation

1. Refuse

Refuse is a general term used to indicate which is rejected or left out as worthless. It may be in liquid, semisolid or solid form.

2. Garbage

It is the dry refuse and includes decayed fruits, grass, leaves, paper pieces, sweepings, vegetables, etc.

3. Sullage

It is the liquid waste from bathrooms, kitchens, washbasin, etc. It is merely a waste water and does not create bad smell.

4. Sewage

It is the liquid waste from the community. It includes sullage, discharge from latrines, urinals, industrial wastes and storm water.

5. Sewer

It is the underground conduit (or) drain through which sewage is conveyed.

6. Sewerage

The entire process of collecting and carrying sewage by water carriage system through sewers is known as sewerage.

7. Invert

It is the lowermost level (or) surface of a sewer.

8. Storm Water

It is the rain water of the locality.

9. Dry Weather Flow

It is the normal flow of sewage during the dry season of the year.

U5.indd 87 12/11/2019 1:03:04 PM